

Lab : Encrypting_and_Decrypting_Message_Using_AES

Objective: To apply the process of encrypting, securely transferring, and decrypting files using AES encryption between Windows Server 2022 and Kali Linux

Tools required: Windows Server 2022 VM, Kali virtual machine

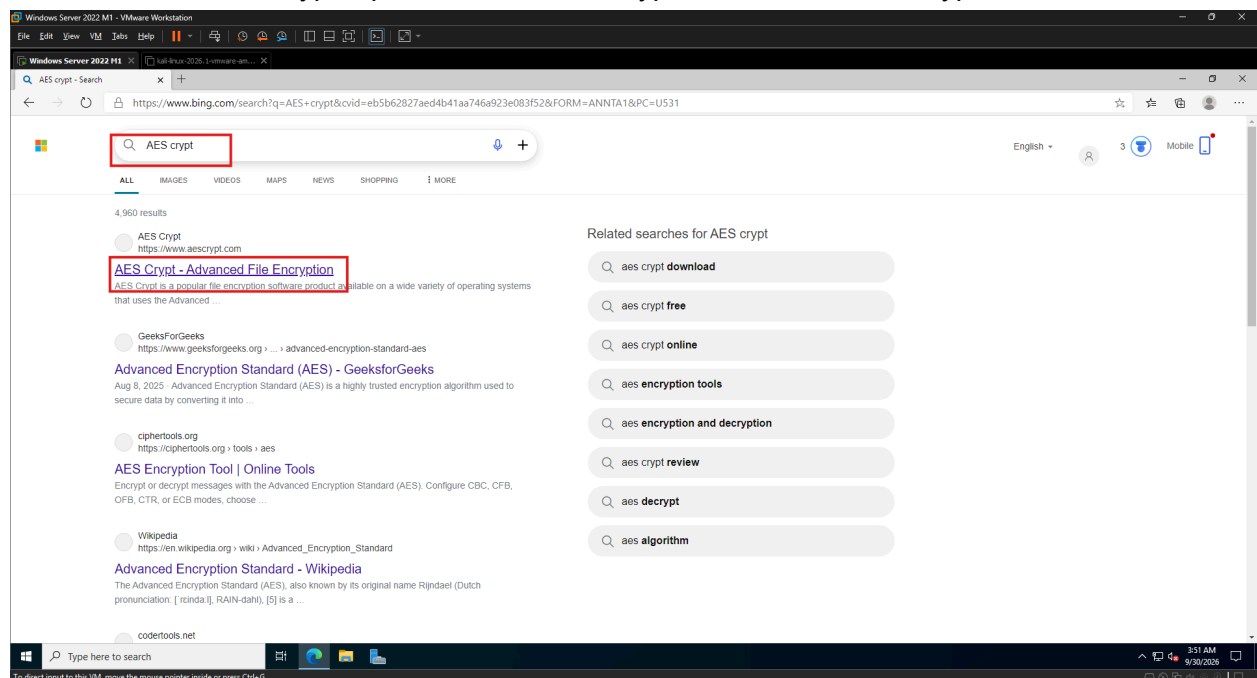
Prerequisites: WinRAR installed on Windows 2022

Steps to be followed:

1. Encrypt the file on Windows Server
2. Transfer the encrypted file to Kali Linux
3. Decrypt the file on Kali Linux

Step 1: Encrypt the file on Windows Server

1.1 Launch Windows Server 2022 VM in the VMware workstation, open Microsoft Edge browser and search for AES Crypt, open link with AES Crypt-Advanced File Encryption



1.2 Download the tool

The screenshot shows the AES Crypt website in a web browser. The navigation menu on the left includes links for Home, Download (highlighted with a red box), Buy, Help, and Information. The main content area features a large banner with the AES Crypt logo and a blue badge for 'New Version 4.7'. Below the banner, there is a section for 'Recommended Download' which lists various download options for Windows, with 'AES Crypt - Desktop & Command-Line (Windows Intel/AMD 64-bit) (Windows 10 or 11)' highlighted by a red box. The page also includes a 'Lifetime License' button and a 'Celebrating 20 Years' badge.

AES Crypt

Advanced File Encryption for Windows, Mac, Linux, BSD, Android, and iOS

Reliable, Trusted, and with Published Source Code

Securing Your Data Since 2005

AES Crypt

AES Crypt is a popular file encryption software product available on a wide variety of operating systems that uses the Advanced Encryption Standard (AES) to easily and securely encrypt files.

The desktop or server version of AES Crypt is available to download for a **30-day free trial**. If you like it, you can buy a perpetual license that gives you access to all future updates. There are no ongoing subscription costs with AES Crypt.

You do not need to be an expert to use AES Crypt, nor do you need to understand cryptography. AES Crypt is designed to be as simple to use as possible, yet still provide the strongest security available. With **over a million downloads**, AES Crypt has made it easy for so many people to secure information.

Using the powerful AES encryption algorithm, AES Crypt can secure your most sensitive files. Once a file is encrypted, you do not have to worry about a person reading your sensitive information, as an encrypted file is completely useless without the password or encryption key.

AES Crypt is the perfect tool for anyone who carries sensitive information with them while traveling, uploads sensitive files to the cloud, stores data at an outside location, sends confidential documents via email, or wishes to protect sensitive information from being stolen from the home or office.

New Version 4.7

Recommended Download[†]

Windows Desktop (Intel/AMD 64-bit)

[†] Based on what your browser claims is your operating system

Windows

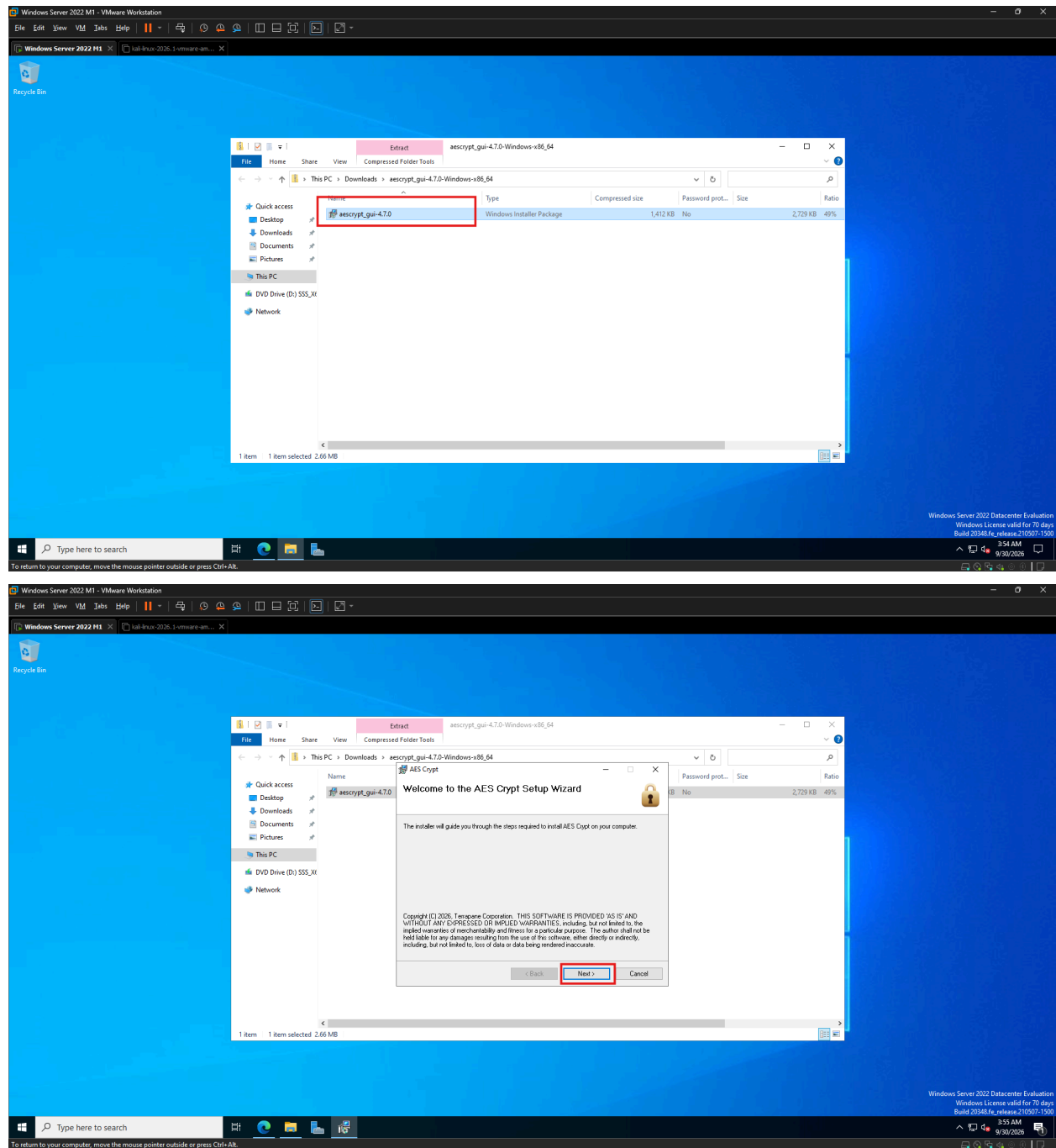
- AES Crypt - Desktop & Command-Line (Windows Intel/AMD 64-bit) (Windows 10 or 11)
- AES Crypt - Desktop & Command-Line (Windows ARM 64-bit) (Windows 10 or 11)
- AES Crypt - Command-Line (Windows Intel/AMD 64-bit) (Windows 10 or 11 stand-alone executable)
- AES Crypt - Command-Line (Windows ARM 64-bit) (Windows 10 or 11 stand-alone executable)

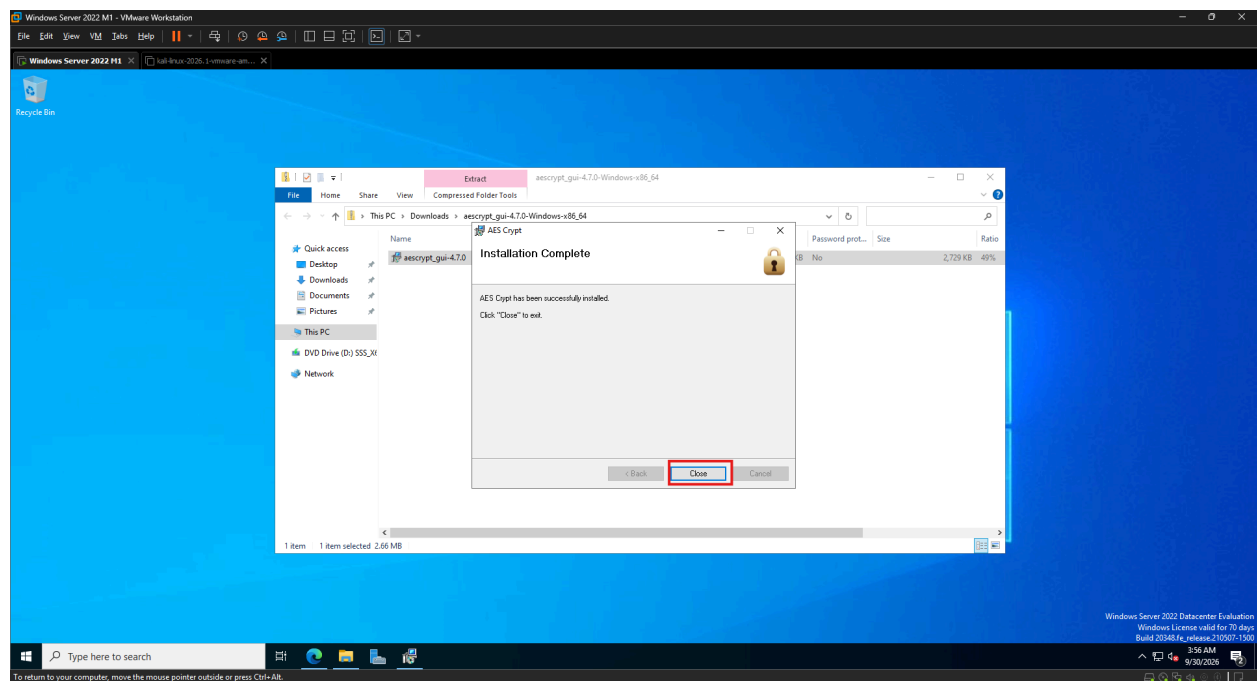
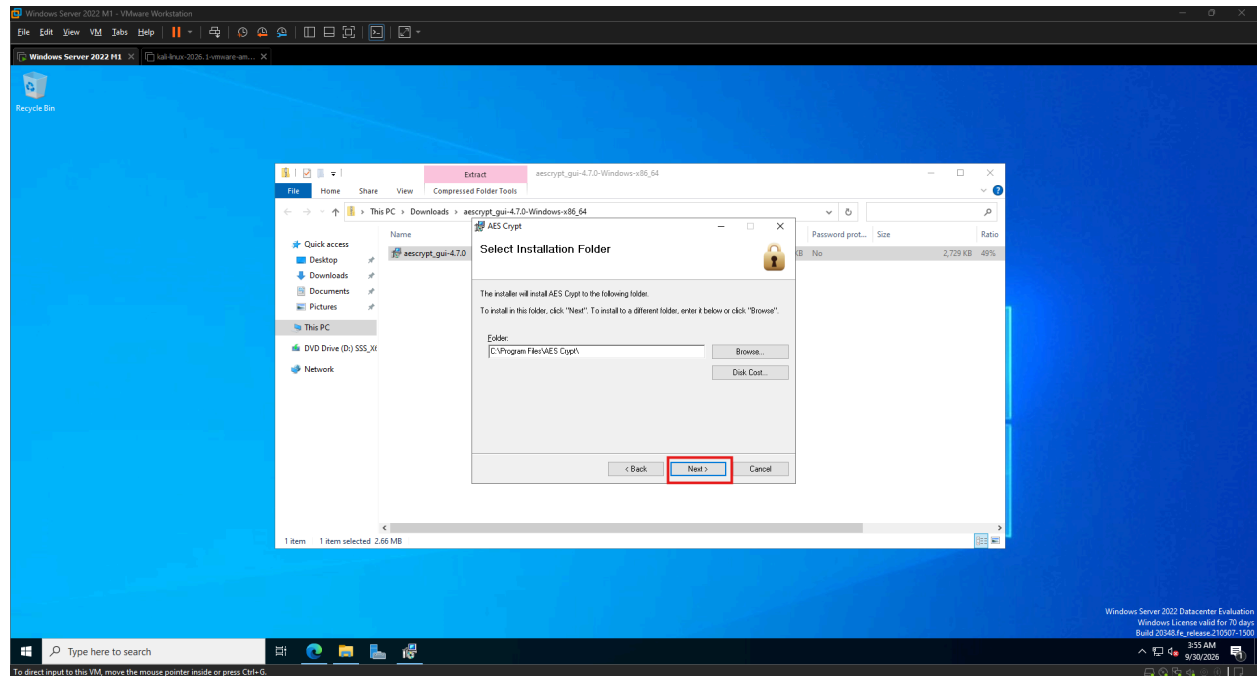
Windows downloads are digitally-signed for your security using a Microsoft-trusted digital certificate.

You may also install the latest version of the AES Crypt desktop version for Windows by typing `winget install Terrapene.AESCrypt` at the command line.

Mac and iOS

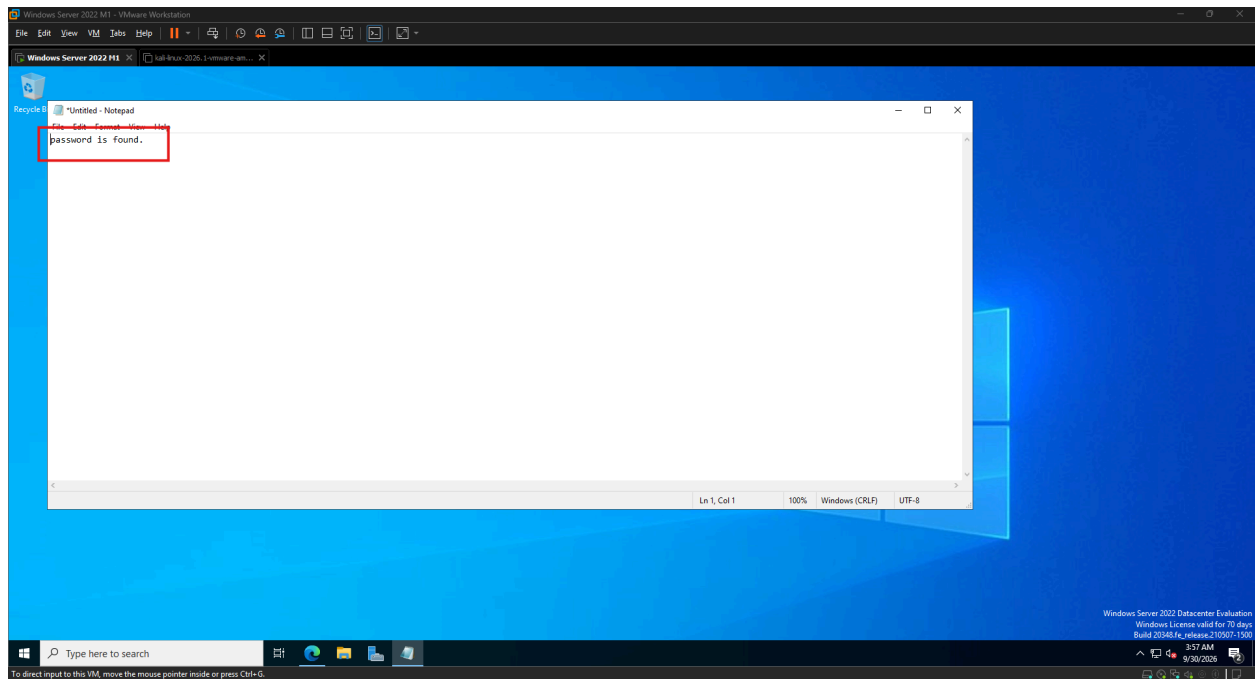
1.3 Extract the file and install it



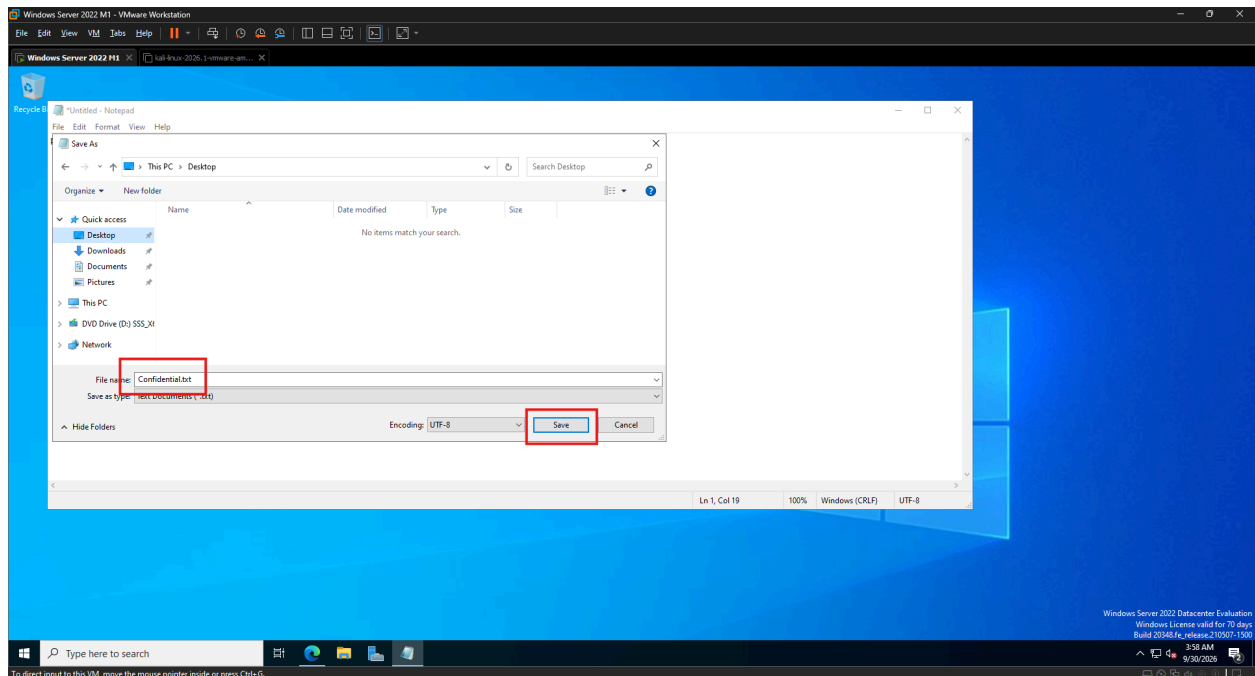


Step 2: Transfer the encrypted file to Kali Linux

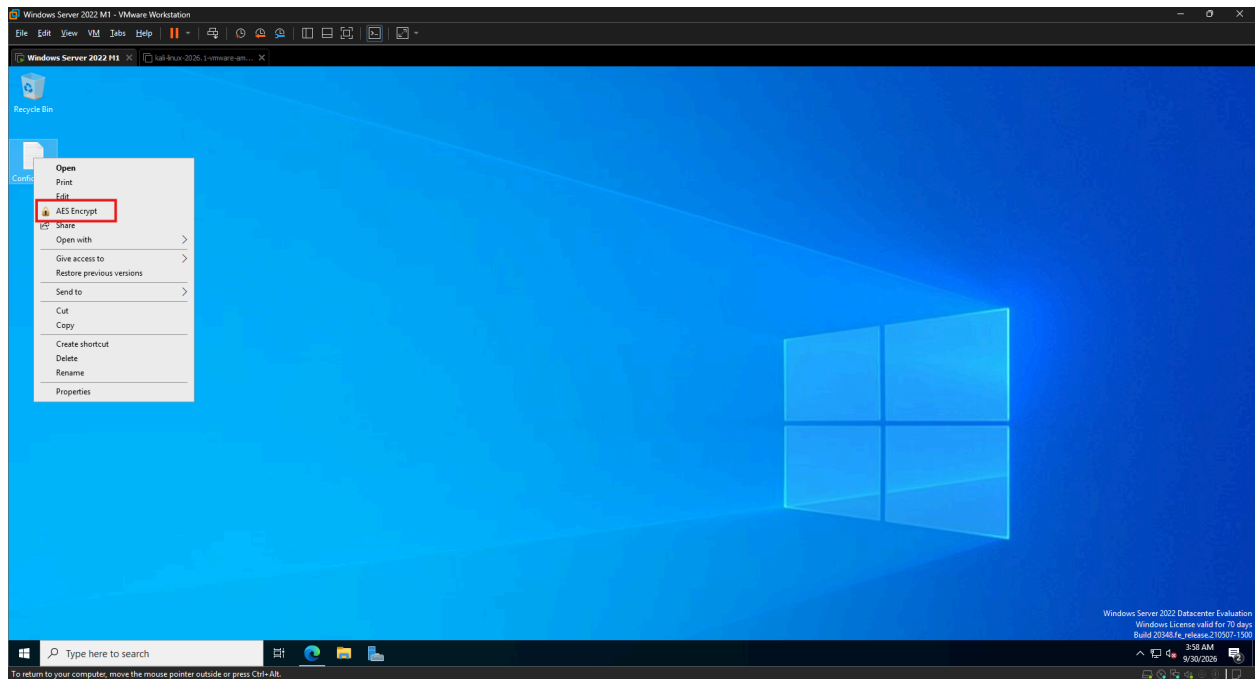
2.1 Open Notepad in Window Server 2022 VM and type the following text inside the Notepad:
password is found.



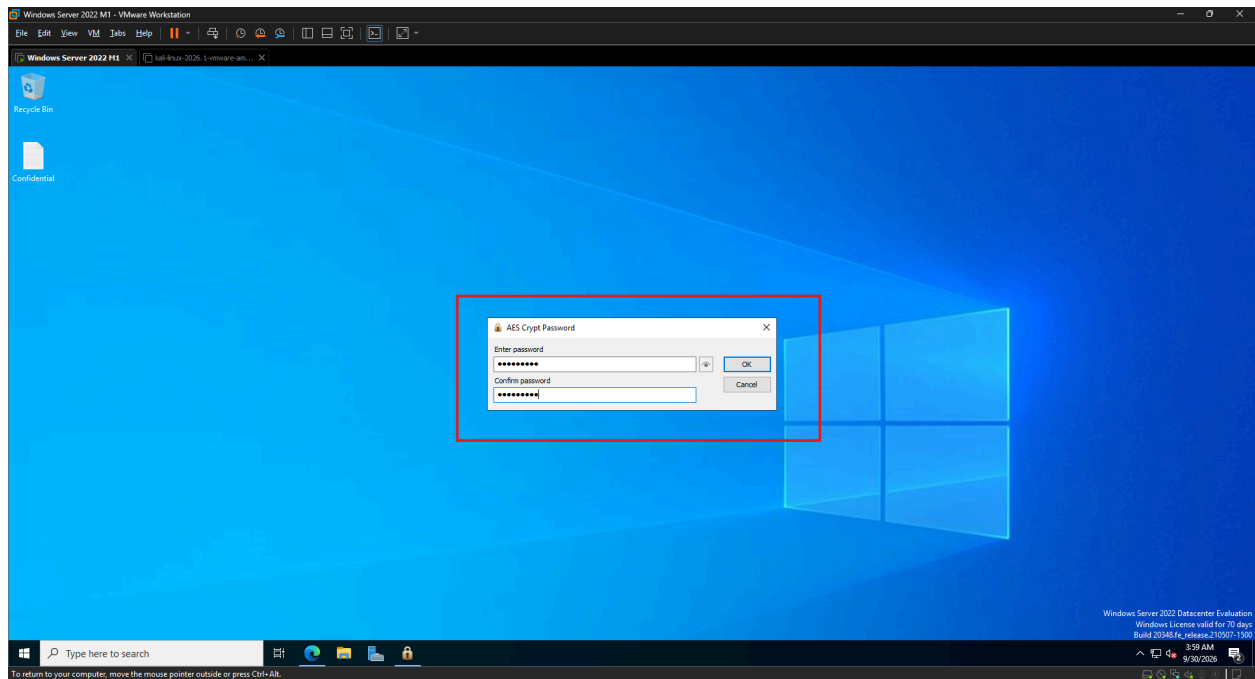
2.2 Name the file ***Confidential.txt*** and save it

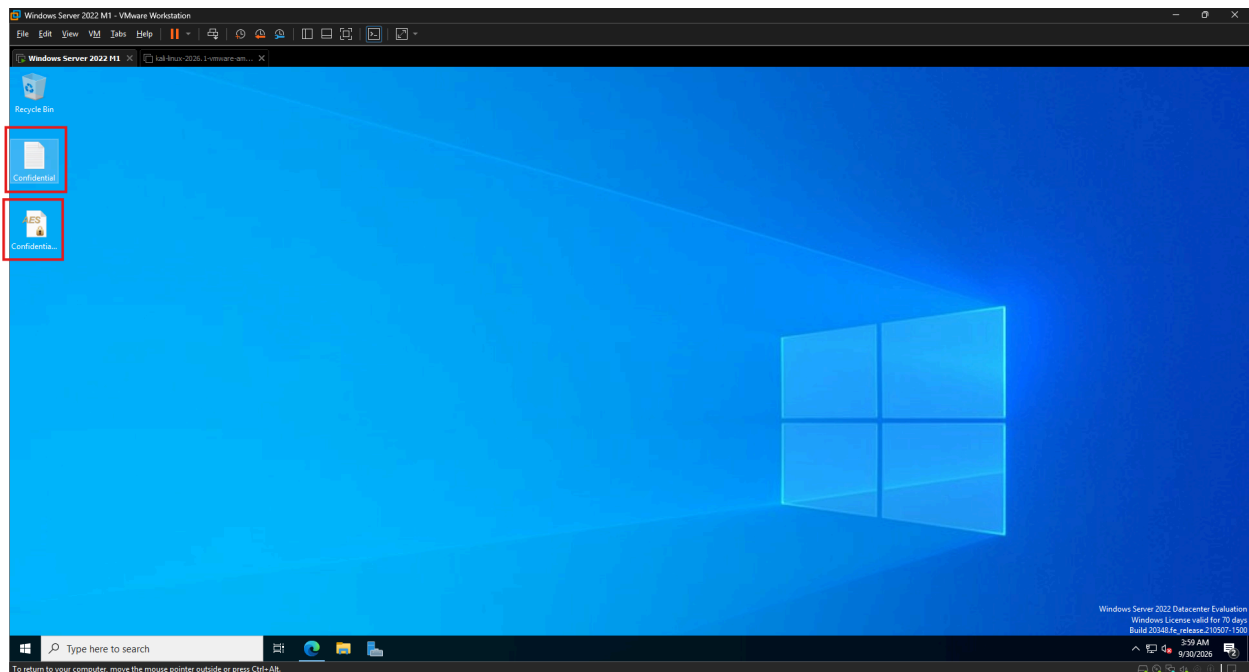


2.3 Navigate to the file **Confidential.txt**, right-click on it & select AES Encrypt



2.4 Enter a password to encrypt the file and select OK





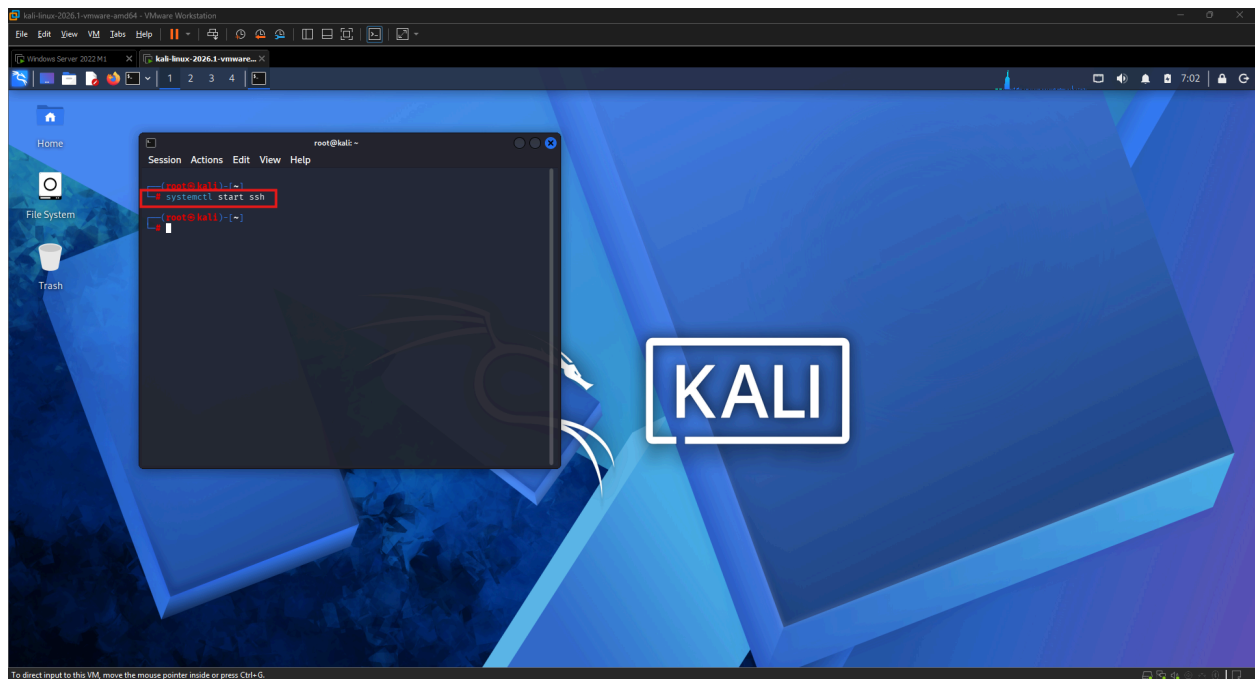
2.5 Open Kali Virtual Machine, launch Terminal Emulator and type the following commands to switch to the root user and check the IP address:

***sudo su –
ip a***



2.6 Type the following command to start the SSH service:

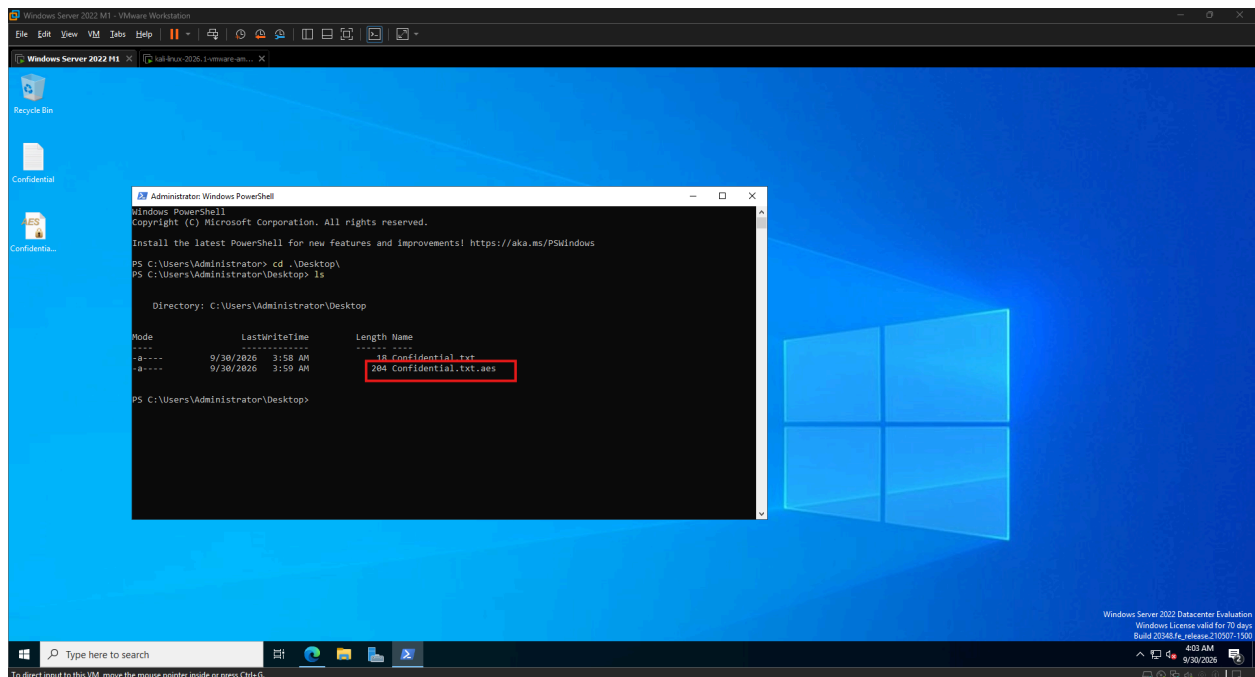
systemctl start ssh



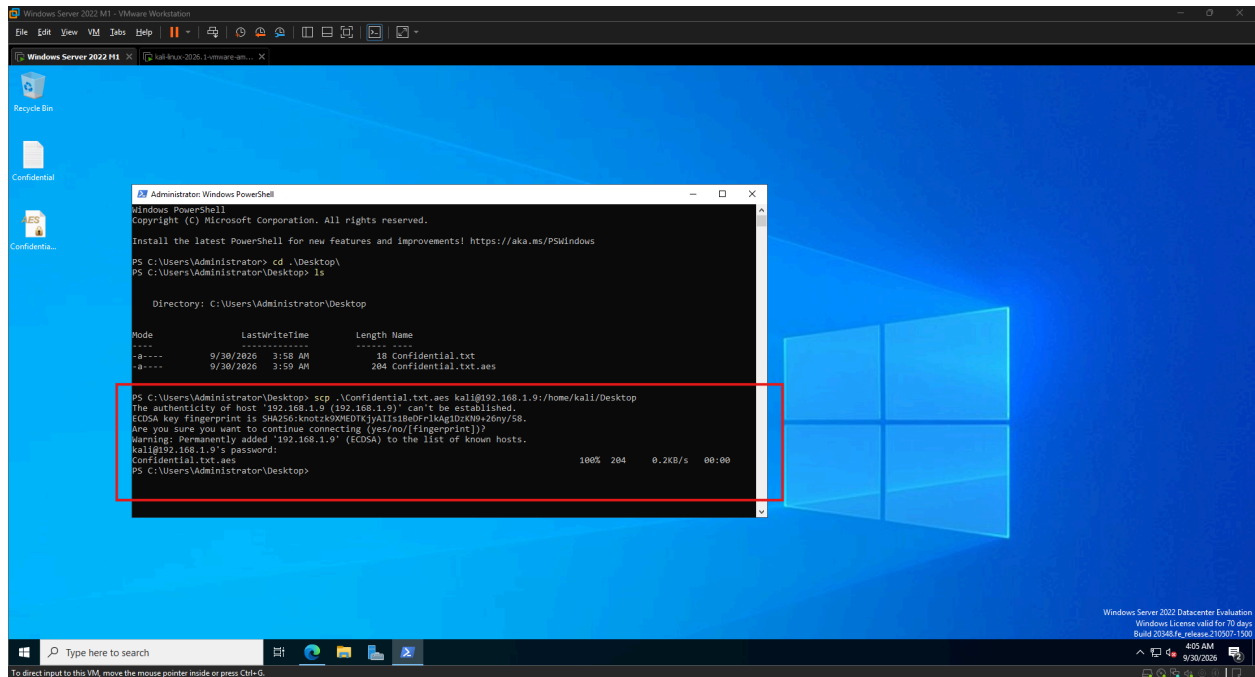
2.7 Navigate back to Windows Server 2022 PowerShell, Type the following commands to navigate to the desktop directory and to view the created files:

cd Desktop

ls

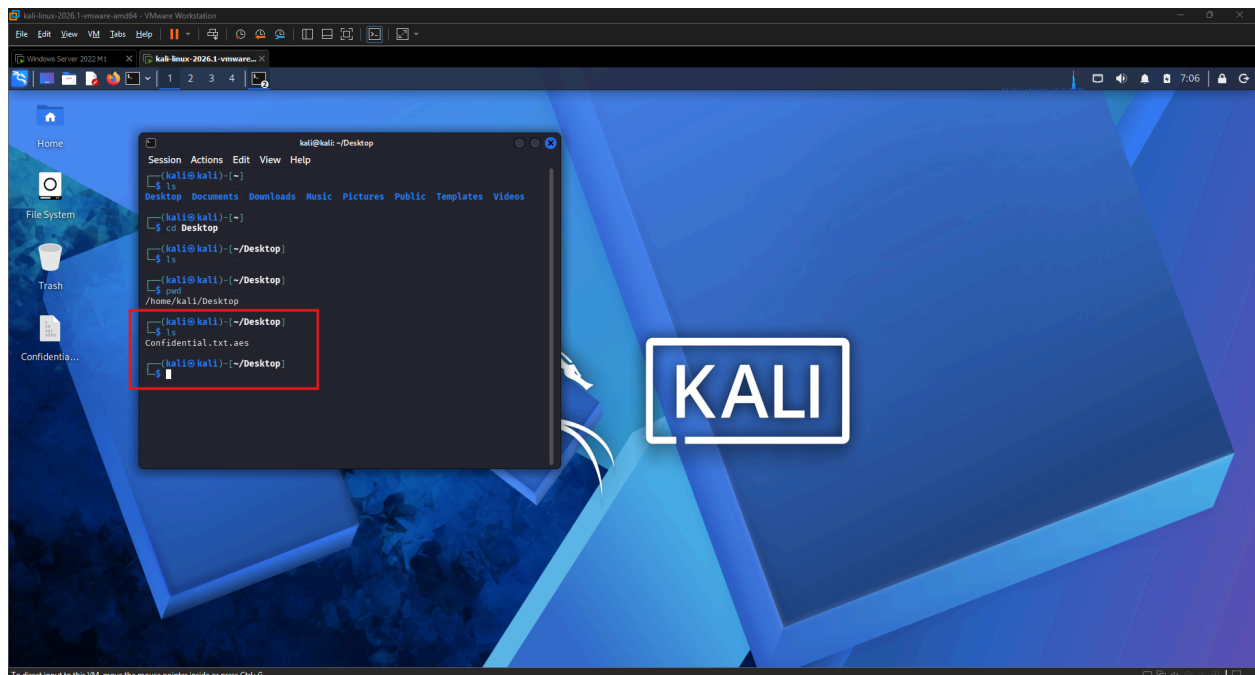


2.8 Type the following command to transfer the file to Kali Virtual Machine :
scp Confidential.txt.aes kali@192.168.1.9:/home/kali/Desktop



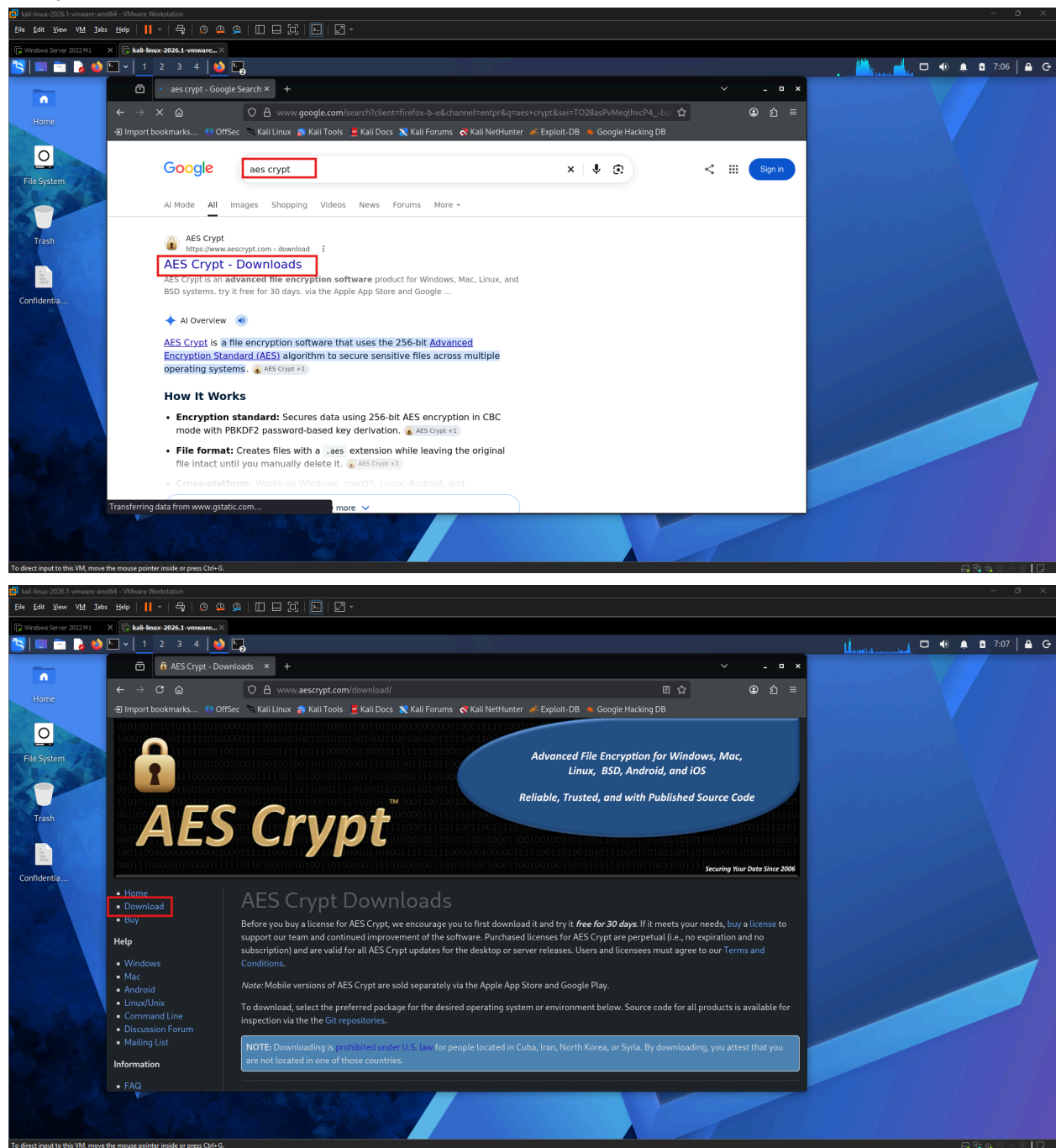
2.9 Navigate back to the Kali virtual machine terminal emulator and type the following command to view the received/copied file:

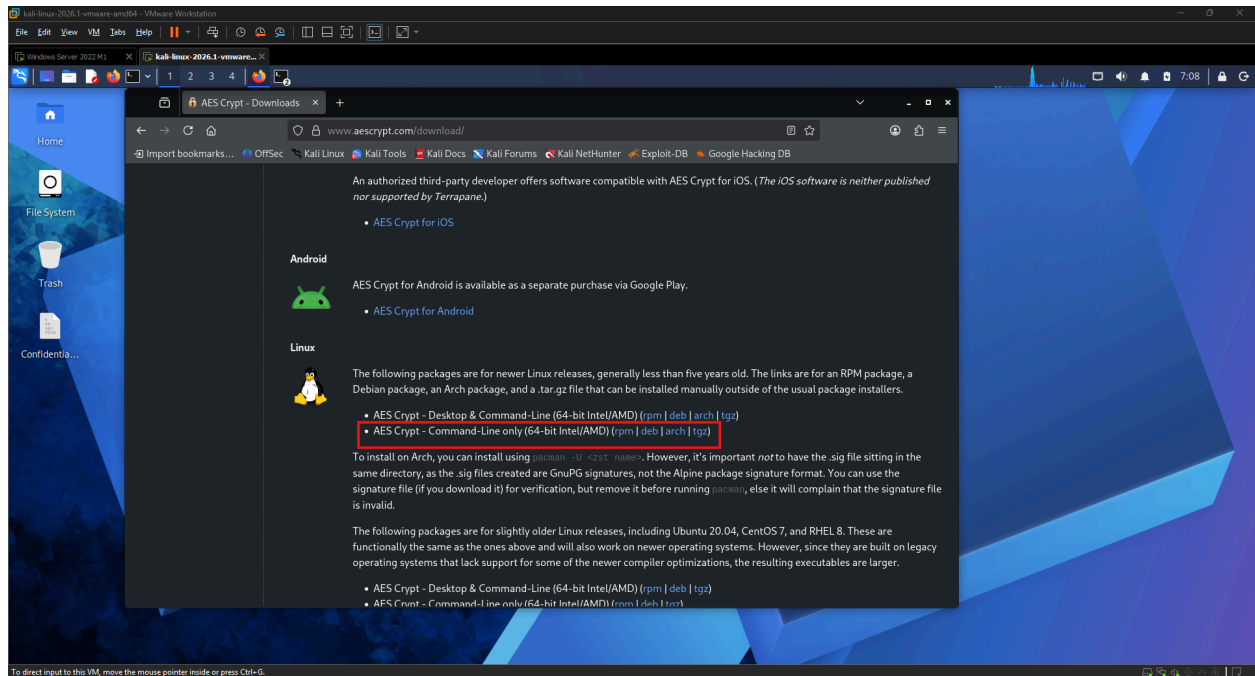
cd Desktop
ls



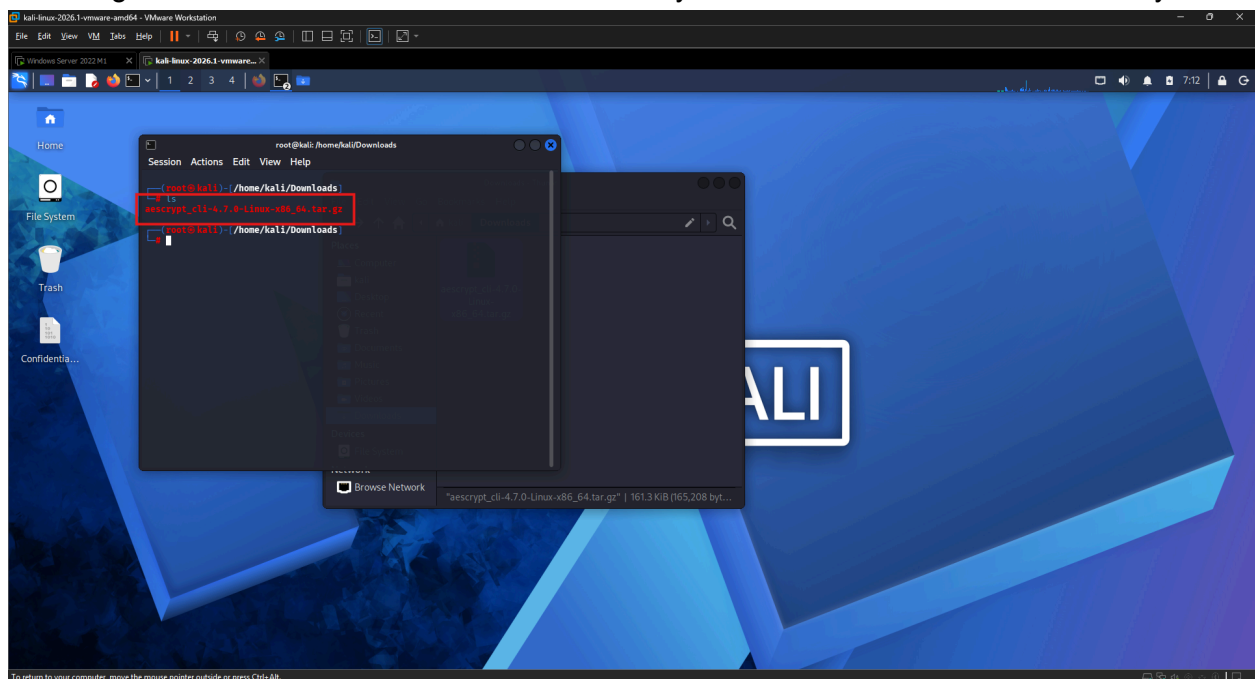
Step 3: Decrypt the file on Kali Linux

3.1 Open Firefox in a Kali Linux virtual machine, download the AES Crypt-Advanced File Encryption



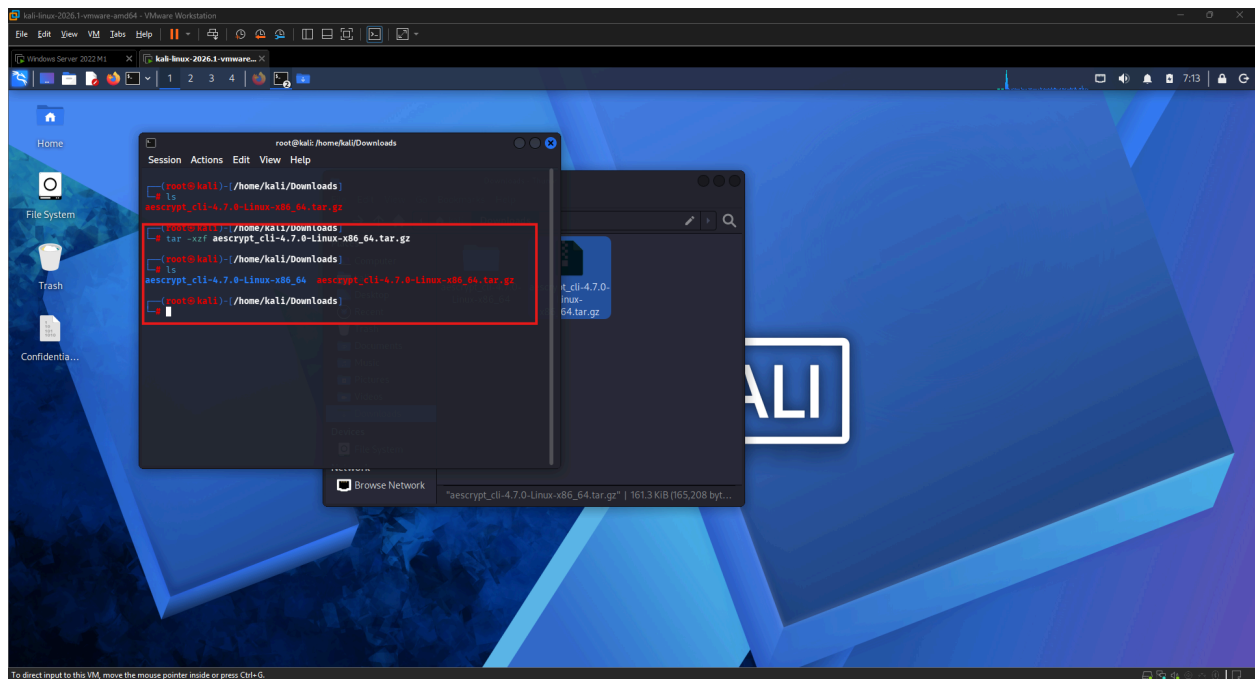


3.2 Navigate back to the terminal emulator and verify if the file is downloaded successfully



3.3 Type the following command to extract the downloaded file:

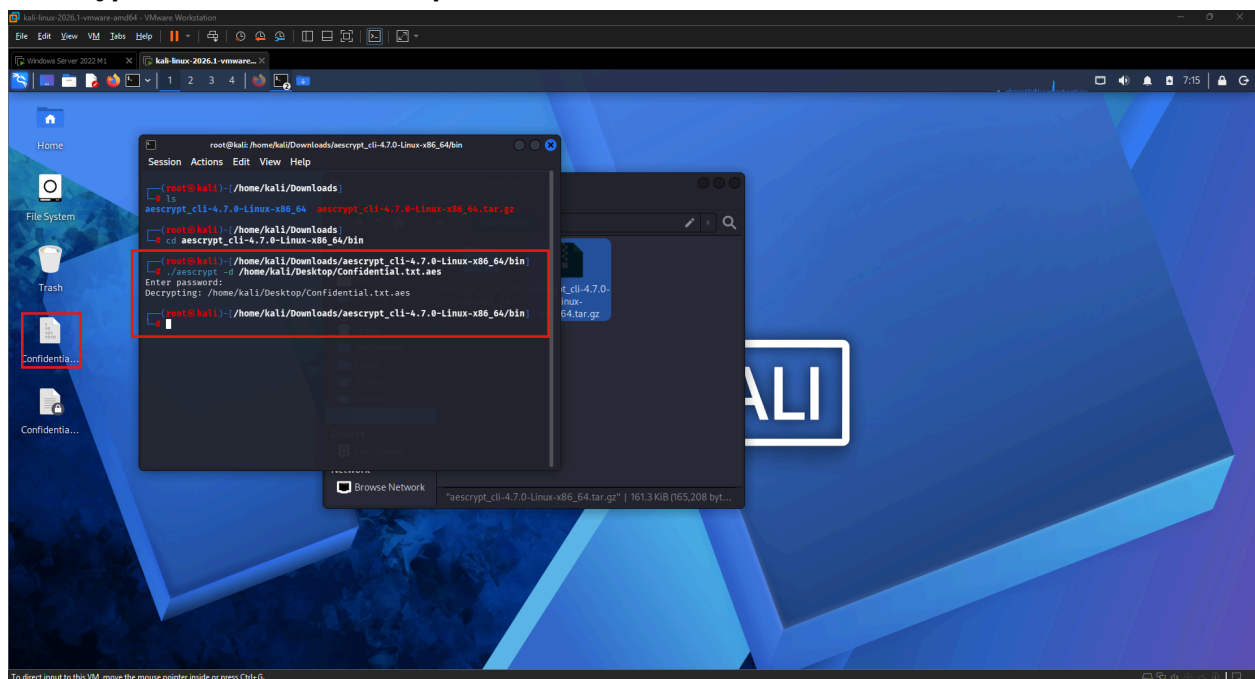
tar -xzf aescrypt_cli_4.7.0-Linux-x86_64.tar.gz



3.4 Type the following command to change the directory to bin and decrypt the file:

cd aescrypt_cli_4.7.0-Linux-x86_64/bin

./aescrypt -d /home/kali/Desktop/Confidential.txt.aes

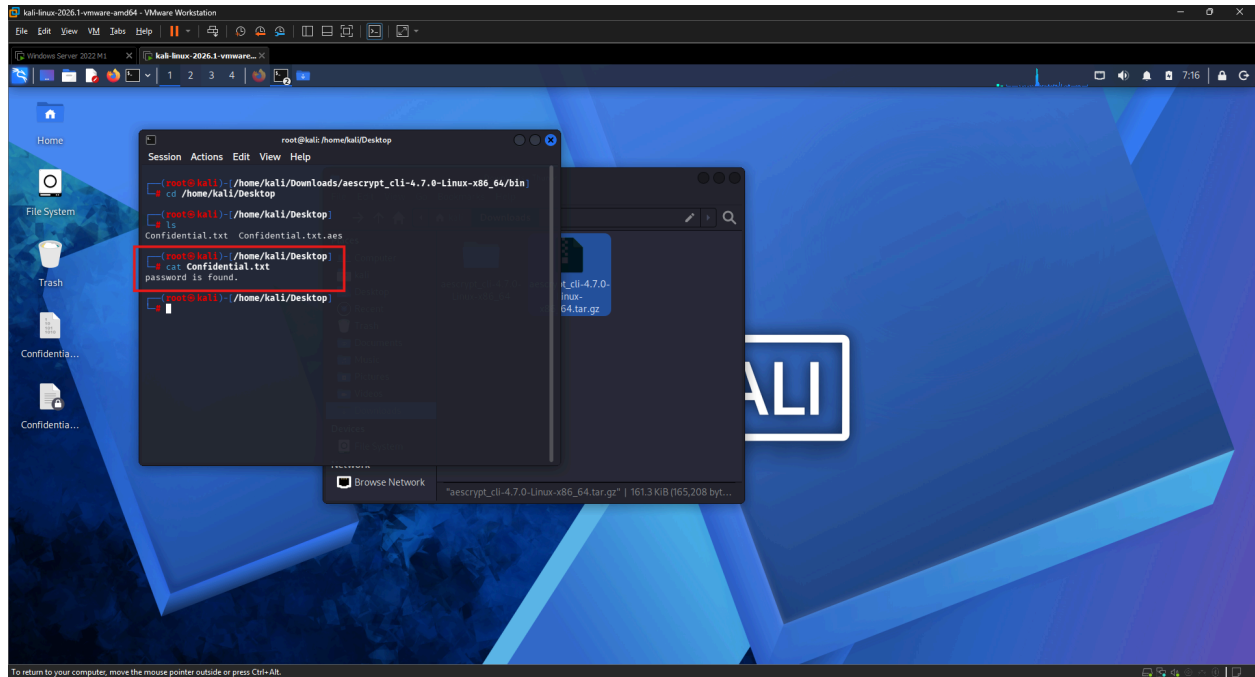


3.5 Type the following commands to view the decrypted file it's content:

`cd /home/kali/Desktop`

`ls`

`cat Confidential.txt`



By following these steps, I have successfully encrypted, transferred, and decrypted a message using AES and secured file transfer across different operating systems.